Digital Mohalla — deployment diagram, iteration 1 (provisional) (design step A13; decision D17; target confirmed by Student 3 by D-14)

UML 2.5 deployment notation: 3-D box = node («device» or «cloud service»); plain box = «executionEnvironment» / «container»; box with document icon = «artifact»; folded note = comment; line = communication path (association) with protocol and port, dashed inside the Docker network. Items marked TBC are confirmed at the D-14 / D-15 gates.

AWS — one region (nearest to the pilot, TBC)

«device»
Mobile phone (Android / iOS) [*]: residents, lenders, moderators and admins

«executionEnvironment»
React Native / Expo runtime (NFR-D-05)

«artifact» **digital-mohalla-app** (one app, screens by role)
lender W1–W2, handover and return W3a–W4, dispute W5, moderator W6–W7 (13_wireframes); EAS build or Expo Go

HTTPS :443
REST /api/v1 (JSON)

«device»
Team laptop: demo fallback

«executionEnvironment» **Docker Compose (local)**
same *docker-compose.yml* and images as production, seed-reset script, synthetic users labelled synthetic. Used if the network fails during the demo; a local run is still live (plan, demo script).

«device»
EC2 instance, Ubuntu 24.04 LTS (size TBC, e.g. 2 vCPU / 2 GB)

«executionEnvironment»
Docker Engine + Docker Compose (*infra/docker-compose.yml*; one internal network)

«container» **proxy**: TLS reverse proxy (Caddy or Nginx)
terminates HTTPS :443 (NFR-S-01); routes */api/v1* to *api*; serves no other content; adds no logging of request bodies

HTTP :8000 (internal network)

«container» **api**: Python 3.12, FastAPI + Uvicorn (modular monolith, A8)
«artifact» *ghcr.io/<org>/dm-api:<commit-sha>*
• Module 1 accounts and locality, Module 3 borrowing lifecycle, Module 5 assurance engine (S2 estimation and policy, S1 indirect trust), Module 9 moderation and audit, Module 7 notify() stub
• «artifact» *Assurance_Engine_Config_Draft.json* (config_version stored on every assessment, NFR-D-01)
• «artifact» migrations from *schema.sql* v2.3, run by CD before start-up
• privacy filter on every response and log line (NFR-PR-01, NFR-S-07)
• *GET /health* for the CD smoke test (NFR-D-04)

SQL / TCP :5432 (internal only)

«container» **db**: PostgreSQL 16 with *btree_gist* (ADR-0002)
«artifact» schema *dm*: event store, projections, bookings (EXCLUDE), assessments, cases, audit log
volume *pgdata*; port 5432 on the internal network only; nightly *pg_dump* (target TBC)

«volume» **attachments**: photos and documents, EXIF location stripped before storage (NFR-PR-01); a host volume in iteration 1, an S3 bucket TBC

Security group: inbound 443 from anywhere; 22 only from the GitHub Actions deploy key and the team; 5432 never public. Secrets come from GitHub Environments into an *.env* file, never the repo.

No coordinates, addresses or CNIC fields exist in any container or volume (schema privacy scan, test suite 14). Notifications are an in-process stub in iteration 1 (no e-mail or push service).

SSH :22
pull, migrate, up -d

HTTPS smoke test
/health + one request

docker pull
(HTTPS)

«cloud service»
GitHub

«executionEnvironment» **GitHub Actions runner**
«artifact» *ci.yml* (every PR and push): ruff, eslint, mypy, bandit, pip-audit, pytest with coverage, integration tests on a Postgres service container, scenario suite, frontend build, JUnit and coverage reports
«artifact» *cd.yml* (push to main): build and push images, deploy over SSH, smoke test, roll back to the previous tag on failure

«artifact» **repository**
backend/, frontend/, infra/, docs/ (this design folder)
protected main and develop; tags v0.1.0-mid1, design-v1.0, schema-v1

docker push

«cloud service»
GitHub Container Registry (GHCR)
«artifact» dm-api:<sha>, dm-proxy:<sha>
previous tag kept for rollback

GitHub Environments: SSH deploy key, database password, any API keys. The handbook forbids committing keys; Dependabot and pre-commit hooks guard the repo.

Why this deployment
• One environment and one host are enough to show real CD for Mid-1 (plan, CI/CD); a staging environment can be added later.
• The backend is one FastAPI process (modular monolith, A8), so one api container serves every module; module boundaries stay code interfaces.
• PostgreSQL is required, not optional: the overlap constraint needs *btree_gist* (ADR-0002).
• Rollback = redeploy the previous image tag after a failed smoke test (NFR-D-03).
• Traceability: NFR-D-02 (clean install from the compose file), NFR-D-03, NFR-D-04, NFR-D-05, NFR-S-01, NFR-PR-01, NFR-S-07.
Rationale and open items: *Deployment_Design.md*.